



Vojtěch Kubáč

SW engineer • C++, web full-stack, applied ML

Experience

2024–today **Software engineer, Ententee**

Full-stack product delivery across healthcare and robotics engagements, developed alongside my role at CFD Support. Stack spans Python (Django, FastAPI), Java / Spring Boot, React / TypeScript, PostgreSQL, with ROS / Gazebo and C++ on the UAV side.

- **DrChrono** • US medical app uplift: Django/GraphQL, React/TS.
- **ALF** • medical-faculty app for attestations: Spring Boot, React/TS, PostgreSQL.
- **Ententee Hub** • internal ops and hiring tooling: FastAPI, React/TS, PostgreSQL.
- **THK UAV** • greenfield stack: perception, mission autonomy, GPS-denied navigation.

2019–today **Software engineer, CFD Support**

Develop and maintain multiple modules of TCAE, a fluid dynamics simulation application deployable on desktop machines or computational clusters. Built around ParaView/VTK and OpenFOAM. Core stack: C++23, Python, CMake, Qt, GitLab CI.

- **TCFD** • Computational Fluid Dynamics module: OpenFOAM, ParaView UI.
- **TFEA** • FEA (solid mechanics) module: CalculiX, CFD coupling, ParaView UI.
- **TCAA** • aeroacoustics: FW-H / Farassat 1A far-field, SPL/octaves.
- **TBASE** • SQLite simulation DB: reproducible archive, ParaView IPC, surrogate modelling.
- **TMESH** • meshing module from STL/STEP (NetGen, Gmsh).
- **TCAE DevOps** • GitLab CI, CTest, headless Qt, Linux/Windows installers.

2019–2020 **PhD researcher, Charles University (MFF UK)**

Research in the group of Christoph Allolio on continuum modelling of lipid membranes.

- Ran molecular dynamics simulations (GROMACS) of lipid membranes.
- Worked on implementation of a continuum mechanics solver in C++ for membrane remodelling.

Education

2017–2020 **Master's, Charles University, Prague, Mathematical Modeling in Physics and Technology**
Thesis: *Comparison of possible formulations of fluid-structure interactions with application in biomechanics*, supervised by RNDr. Jaroslav Hron, Ph.D.

2018–2019 **Erasmus, Technical University of Munich, Germany, Faculty of Mathematics**
Machine learning, deep learning, high-performance computing.

2014–2017 **Bachelor's, Charles University, Prague, General Mathematics**

Skills

Programming	C++ (C++23), Python, TypeScript, Java, Bash
Platform	Qt, CMake, CTest, SQLite, GitLab CI, NSIS, Linux, Git
AI / ML	RAG / LLM integration; surrogate modelling over parametric design spaces
Scientific	OpenFOAM, CalculiX, NetGen, Gmsh, ParaView, VTK, FEniCS, GROMACS
Web BE	Django, Spring Boot, FastAPI, GraphQL, PostgreSQL
Web FE	React, Svelte, Tailwind CSS, Vite
Robotics	ROS, Gazebo
Spoken	Czech (native), English (C1), German (B1)

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